Distance sampling, step by step

C. H. Ross. 2026-11-05. Distributed with the package source.

1. Scope

The steps from a NARWC survey file to an abundance estimate, each with the code that runs it. Every code block was run, in order, on the package fixture (`inst/extdata/narwc-example.csv`). The numbers quoted are from that run.

For what each function does internally, see `docs/10-code-guide.md`.

Section	Function	Package
2. Read	`read_narwc()`	narwcr, re-exported
3. Prepare	`prepare_survey()`	distsamp
4. Check	`validate_narwc()`, `diagnose_pipeline()`	distsamp
5. Segment	`segment_survey()`	distsamp
6. Inspect	`segs$segments`, `segs$detections`	distsamp
7. Detection table	`detection_data()`	distsamp
8. Detection function	`ds()`	Distance
9. Density surface inputs	`segments_as_sf()`	distsamp, sf
10. Density surface	`dsm()`	dsm
11. Abundance	`predict()`, `dsm_var_gam()`	dsm

2. Read

```
library(distsamp)
path <- system.file("extdata", "narwc-example.csv", package = "distsamp")
dat  <- read_narwc(path)
```

Column names are standardised and the mapping is printed. Result: 113 records.

3. Prepare

```
dat <- prepare_survey(dat, platform = "aerial")
```

In order: line occupations (`LEGNO3`), line state filled (`LEGSTAGE`), platform filter by speed, effort flags (`OnOff.Effort`). Result: 101 of 113 records on effort.

The order is fixed. Filtering the platform before line occupations merges occupations and counts the ferry between them as effort.

4. Check

```
findings <- validate_narwc(dat)
diagnose_pipeline(dat, seg_length = 5)
```

`validate_narwc()` returns one row per rule broken (2 on the fixture). `diagnose_pipeline()` reports, stage by stage, what each step kept and why. Run both before trusting any number from a new extract (`docs/08-onboarding-a-real-extract.md`).

5. Segment

```
segs <- segment_survey(dat, seg_length = 5, seed = 1)
```

On-effort track is cut into segments of about 5 km. Steps of Section 3 that have not run are run here, except the platform filter. Perpendicular distances are attached to sightings. Circling sightings are added to the group they were counted with. `seed` makes the cut reproducible.

Main options:

Argument	Default	Effect
`seg_length`	required	target segment length, km
`seed`	`NULL`	reproducible segmentation
`species`	all	restrict sightings
`distance_units`	`"m"`	units of `distance`
`circling`	`"same_species"`	which circling sightings are kept
`circling_distance`	`"with_group"`	circled animals join their group; `"break_off"` makes them separate detections
`circling_max_gap_km`	`seg_length`	furthest a group may be from the break-off

Result: 20 segments, 92.23 km of effort (all of it), 8 detections.

6. Inspect

```
segs$segments      # one row per segment: seg_id, seg_eff (km), mid_lat, mid_lon
segs$detections    # one row per sighting: seg_id, size, distance, distance_source
table(segs$detections$distance_source)
```

`distance_source` names where each distance came from: `angle`, `exact`, `strip`, or for circling sightings `circling/break_off`. On the fixture all 8 are `angle`.

7. Detection table

```
flat <- detection_data(segs, area = 5811)
```

One row per detection, plus one row per segment with no detection, in the shape `Distance` expects. `area` is the study area in km²; it has no default. Circling detections are left out by default.

Exact distances give a `distance` column; `STRIP` bins give `distbegin` and `distend`. Never both: `ds()` reads the bin columns as binned data.

8. Detection function

```
library(Distance)
fit <- ds(flat, key = "hn", adjustment = NULL, truncation = "10%",
          convert_units = 0.001)
summary(fit)
```

`convert_units = 0.001` because distances are in m and effort in km. Leaving it out scales abundance by 1000.

Result: half-normal, average detection probability 0.772, 7 detections after truncation. Abundance from the design-based estimate: 936.6 in 5,811 km².

This assumes detection on the line is certain, `g(0) = 1`. For aerial whale surveys it is not (`docs/07-fitting-architecture.md`, Section 5).

9. Density surface inputs

```
pts <- segments_as_sf(segs, "midpoints")
aea <- "+proj=aea +lat_1=40 +lat_2=46 +lat_0=43 +lon_0=-69 +datum=WGS84 +units=km"
xy  <- sf::st_coordinates(sf::st_transform(pts, aea))

segment.data <- data.frame(
  Sample.Label = segs$segments$seg_id,
  Effort       = segs$segments$seg_eff,
  X            = xy[, 1],
  Y            = xy[, 2]
)

used <- fit$ddf$data
observation.data <- data.frame(
  object       = used$object,
  Sample.Label = used$Sample.Label,
  size         = used$size,
  distance     = used$distance
)
```

Midpoints are projected to an equal-area projection in km before modelling. Observations are taken from the detections the detection function was fitted to (`fit$ddf$data`), so truncated detections are excluded and `object` matches.

10. Density surface

```
library(dsm)
mod <- dsm(count ~ s(X, Y, k = 5),
           ddf.obj = fit,
           segment.data = segment.data,
           observation.data = observation.data,
           family = tw(), method = "REML",
           convert.units = 0.001)
summary(mod)
```

Count per segment is modelled as a smooth of position, with the segment's searched area as offset. `convert.units = 0.001` for the same reason as in Section 8. `k = 5` only because the fixture has 20 segments; a real survey supports more.

Result: 22.1% deviance explained.

11. Abundance

```r
grid <- expand.grid(
  X = seq(min(xy[, 1]), max(xy[, 1]), by = 5),
  Y = seq(min(xy[, 2]), max(xy[, 2]), by = 5)
)
grid$off.set <- 25                                    # cell area, km^2
grid$N <- predict(mod, newdata = grid, off.set = grid$off.set,
                  type = "response")
sum(grid$N)

v <- dsm_var_gam(mod, pred.data = grid, off.set = grid$off.set)
summary(v)
```

Abundance is predicted on a grid of 5 × 5 km cells and summed. The grid here is the bounding box of the segments; a real analysis predicts over the study area, with the same covariates the model used.

Result: 99 cells, 352.4 animals, CV 0.61. This differs from Section 8 because it covers only the bounding box, not the 5,811 km² study area.

12. Units

Quantity	Unit
`seg_eff`, `Effort`	km
`distance`	m (default) or km
`area`, `off.set`	km²
`convert_units`, `convert.units`	0.001 for m distances and km effort; 1 for km and km

13. Limitations

- The fixture is synthetic and small. The fits above show the code runs; the estimates mean nothing.
- `g(0) = 1` is assumed throughout. Availability and perception bias are not corrected.
- Detection-function model selection (keys, adjustments, covariates) belongs to the dsfit package, and project choices to the dsmfit repository. This chapter fits one model of each.

14. Verification performed

Check	Result
Every code block, run in order on the fixture	Runs without error. Numbers above are from that run.
`ds()` on a flatfile of exact distances	Failed before this date: `detection_data()` emitted all-NA `distbegin`/`distend`, which `ds()` reads as binned data. Corrected; test added in `tests/testthat/test-detection-data.R`.
Vignette `from-segments-to-density.Rmd` fitting code	Not run by `R CMD check` (`eval = FALSE`). Missing `convert_units` in `ds()` and `convert.units` in `dsm()`, and referred to an undefined `df_data`. Corrected to the code in Sections 8–10.

15. References

Buckland, S.T., Anderson, D.R., Burnham, K.P., Laake, J.L., Borchers, D.L. and Thomas, L. (2001) *Introduction to Distance Sampling.* Oxford University Press.

Miller, D.L., Burt, M.L., Rexstad, E.A. and Thomas, L. (2013) Spatial models for distance sampling data: recent developments and future directions. *Methods in Ecology and Evolution* 4(11):1001–1010.

Miller, D.L., Rexstad, E., Thomas, L., Marshall, L. and Laake, J.L. (2019) Distance sampling in R. *Journal of Statistical Software* 89(1):1–28.

Kenney, R.D. (2023) *The North Atlantic Right Whale Consortium Database: A Guide for Users and Contributors, Version 8.* NARWC Reference Document 2023-01.